



Economic Insights

Economics - North America

US: Skewed vision – evaluating trimmed-mean inflation

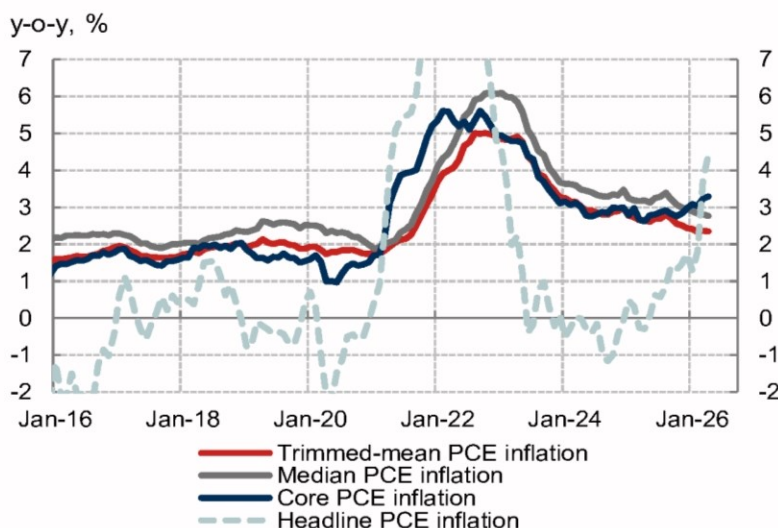
- Fed Chair Kevin Warsh described core PCE inflation, the Fed's long-time preferred inflation metric, as a "rough swag" and proposed alternative inflation measures including trimmed-mean inflation.
- PCE trimmed-mean inflation tends to be slow in detecting a change in the inflation trend, while underlying inflation trend measures indicate the risk of a new upward trend emerging recently.
- We remain skeptical about the usefulness of PCE trimmed-mean inflation, whose method of removing outliers might not be optimal due to the recent shift in skewness of the cross-sectional inflation distribution. Trimmed-mean inflation might underestimate underlying inflation by about 48bp on a y-o-y basis.
- A primary factor making PCE trimmed-mean inflation negatively biased is its failure to capture changes in goods price inflation dynamics in the post-pandemic era.

What is PCE trimmed-mean inflation?

Monthly inflation data are noisy, and idiosyncratic moves by a few components can have an outsized effect. Policymakers aim to look through temporary noise and gauge the underlying trend of inflation. One of the simplest metrics that diminishes such noise is core PCE inflation, which excludes volatile energy and food prices. Most FOMC participants put a heavy weight on PCE core inflation as a gauge of the underlying trend.

Fed Chair Kevin Warsh has criticized PCE core inflation, describing it as a "rough swag" at his confirmation hearing. Instead, he proposed alternative inflation measures, including PCE trimmed-mean inflation and private sector price measures.

Fig. 1: The 12-month change in PCE trimmed-mean stands at 2.35% in April, 94bp lower than 3.29% for core PCE inflation



Source: BEA, Dallas Fed, Cleveland Fed, Haver, Nomura

PCE trimmed-mean inflation is constructed from the cross-sectional distribution of monthly price changes and is designed to remove components with large monthly swings in either direction. Currently, the metric – published by the Dallas Fed – trims 24% from the lower tail and 31% from the upper tail of the distribution. The 12-month change in PCE trimmed-mean stood at 2.35% in April, 94bp lower than 3.29% for core PCE inflation (*Fig. 1*).

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Tracking “true inflation”

A key factor in evaluating the value of trimmed-mean inflation is whether it accurately captures “true inflation,” or underlying inflation. In order to assess whether PCE trimmed-mean inflation captures changes in the inflation trend in a timely manner, we calculate four “true” inflation trend series using ex-post data;

- A centered 36-month moving average of annualized monthly headline PCE inflation
- A band-pass filtered annualized monthly headline PCE inflation
- The average of annualized monthly PCE inflation for the contemporaneous and following 24 months
- A centered 12-month moving average of a centered 24-month moving average of annualized headline PCE inflation

Among these four “true” inflation measures, *a Dallas Fed economist* used the first three to choose the optimal cutoffs when formulating PCE trimmed-mean inflation. Note that the optimal trimming proportions (24% of the lower tail and 31% of the upper tail) were determined to minimize the gap between PCE trimmed-mean inflation and the average of those true inflation measures over the period of January 1977 to June 2009. The last “true” inflation measure was a benchmark *Cleveland Fed economists* used to assess the usefulness of not-seasonally adjusted Median CPI inflation.

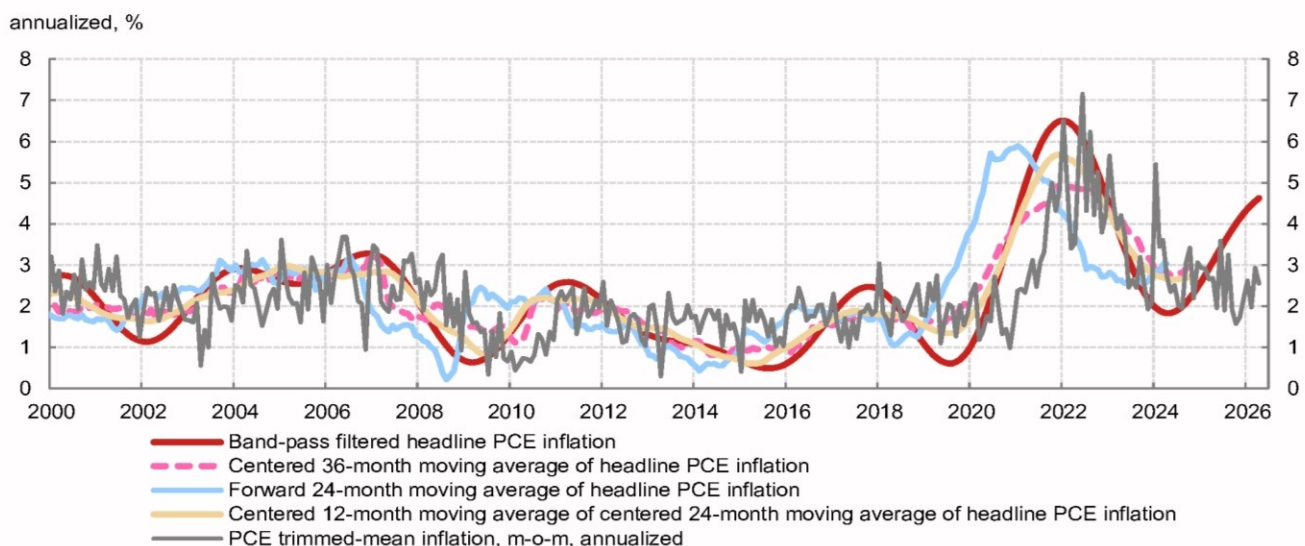
Evaluating trimmed-mean's performance

PCE trimmed-mean failed to detect the post-pandemic inflation surge

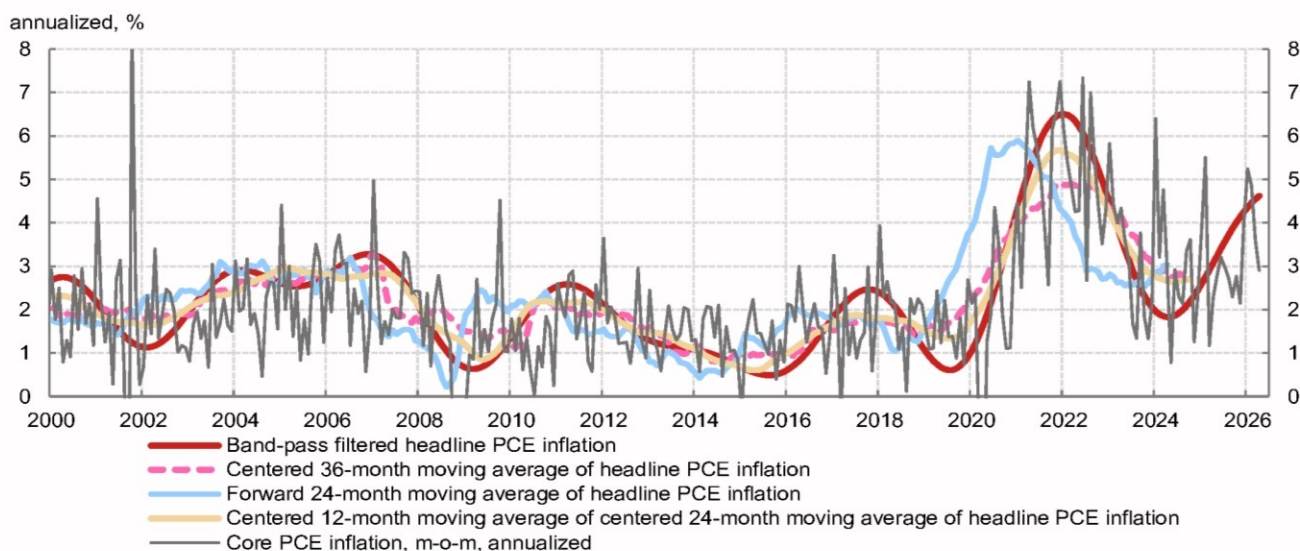
PCE trimmed-mean inflation is designed to look through idiosyncratic noise and avoid sending false positive signals about the underlying inflation trend. Thus, PCE trimmed-mean inflation is often better than core PCE inflation in the sense of fewer false signals of new inflation trends. However, that same feature can make it less responsive when a new inflation regime is emerging, especially if the initial shock is concentrated in a narrow set of components, leading to a change in skewness of the inflation distribution.

This limitation was evident during the post-pandemic inflation surge. At the onset of post-pandemic inflation acceleration, annualized monthly PCE trimmed-mean inflation lagged these four true inflation measures (*Fig. 2*). On the other hand, annualized monthly core PCE inflation started accelerating at almost the same time as the true measures, except for the average of future 24-month inflation (*Fig. 3*).

Fig. 2: PCE trimmed-mean inflation lagged four “true” inflation measures in 2021-2022



Source: BEA, Haver, Nomura

Fig. 3: Core PCE inflation has tracked four "true" inflation measures better than PCE trimmed-mean inflation in recent years

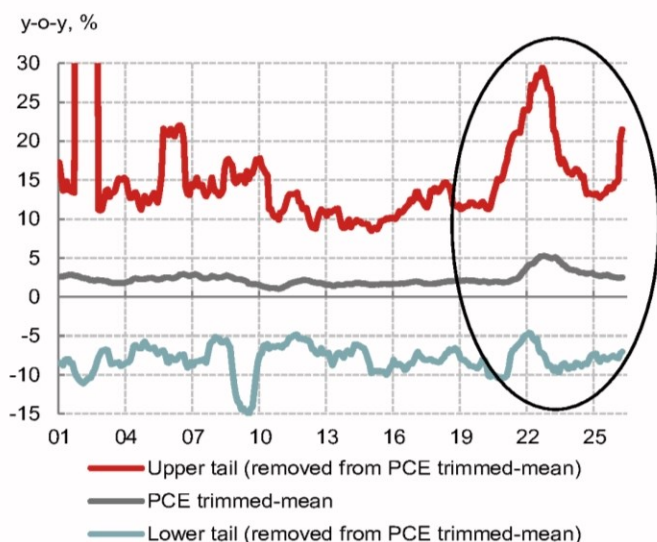
Source: BEA, Haver, Nomura

A similar dynamic appears to have played out in the opposite direction during the post-GFC disinflation. PCE trimmed-mean inflation was slow to respond to the emerging downshift in underlying inflation, while core PCE inflation began to decelerate earlier.

The components excluded from PCE trimmed-mean may offer a better early signal around inflation turning points. A closer look at the distribution of cross-sectional monthly price changes suggests the weighted average of components trimmed from both tails began to react to the pandemic-induced inflation wave as early as Q1 2021, much earlier than PCE trimmed-mean inflation did (Fig. 4). The same dynamic was visible in the opposite direction during the post-GFC disinflation, when components removed from the lower tail dropped sharply, while PCE trimmed-mean inflation remained steady for some time.

Fig. 4: Components removed from PCE trimmed-mean inflation reacted quickly to the post-pandemic acceleration

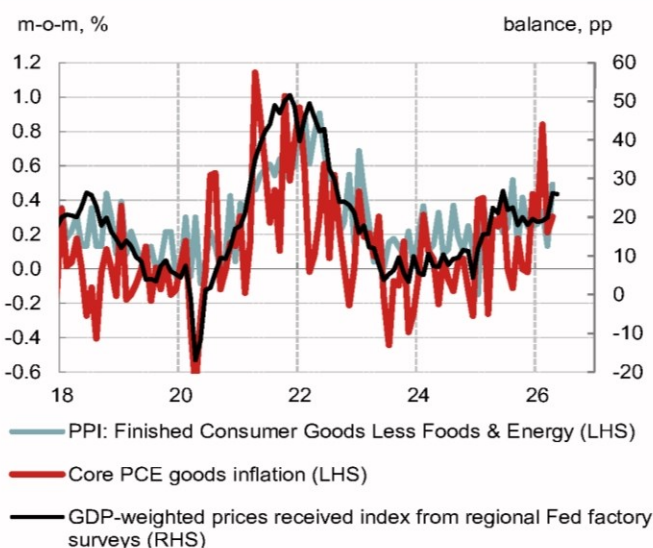
Weighted average of the upper and lower tails removed from PCE trimmed-mean inflation vs. PCE trimmed-mean inflation



Source: BEA, Haver, Nomura

Fig. 5: Manufacturers seem to be passing higher costs on to their customers

Business surveys' prices received indices and core goods inflation



Source: BLS, BEA, NY Fed, Philly Fed, Dallas Fed, KC Fed, Haver, Nomura

Recent data show a similar reason for caution. The upper tail of the distribution has picked up, driven by higher tariffs and energy prices, and remains significantly above the pre-pandemic average (2000-2019). In addition, the lower tail also appears to be rising slightly. Given the post-pandemic experience, as well as recent signs of pass-through of

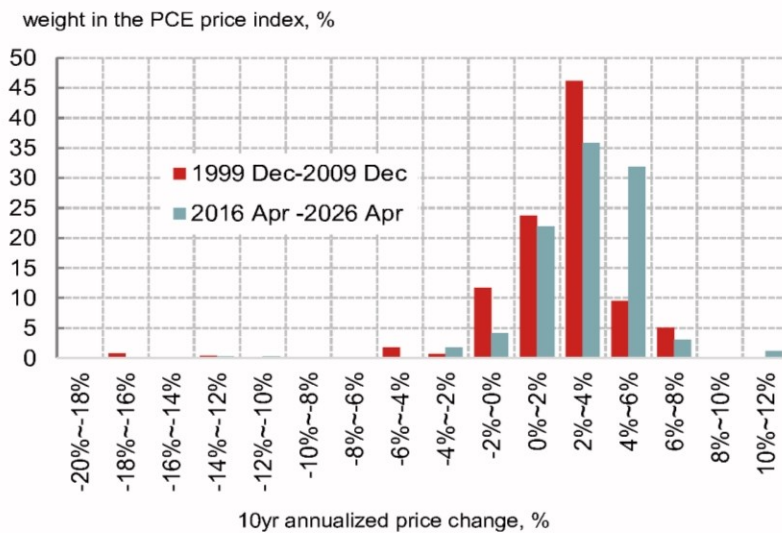
higher costs into consumer goods prices (which is evident in the recent firming of business surveys' prices received indices) (Fig. 5), the recent deceleration in PCE trimmed-mean inflation is unlikely to make policymakers comfortable.

A shift in skewness caused trimmed mean to understate underlying inflation

Another issue with PCE trimmed-mean inflation is that it appears to underestimate underlying inflation. Since the *last update of trimming specifications* by the Dallas Fed in 2009, the distribution of the cross-sectional monthly price changes has shifted, which has likely made the current trimming proportions suboptimal. The distribution of cross-sectional price changes over the past ten years looks to have shifted up (Fig. 6).

Fig. 6: The distribution of cross-sectional price changes appears to have shifted up in recent years

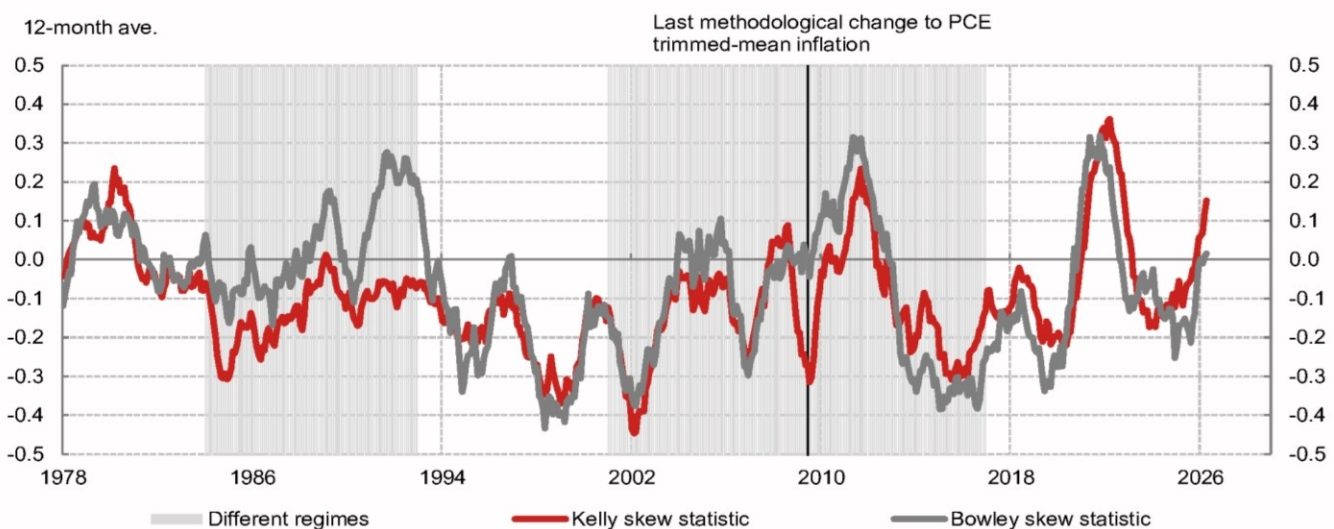
Distribution of cross-sectional annualized price changes



Source: BEA, Haver, Nomura

Using monthly price changes of 178 subcomponents of the PCE price index with their corresponding weights, we calculate two typical skewness measures: the Bowley and Kelly coefficients, which focus on middle 50% and middle 80% of the data, respectively. We take the one-sided 12-month moving average of skewness scores for the monthly price change distribution (Fig. 7). Both measures should fall between -1.0 and +1.0 with a negative (positive) reading, indicating a negatively (positively) skewed distribution.

Fig. 7: Since the last methodological update in 2009, the distribution of cross-sectional price changes has become positively skewed more often



Note: *Regimes refer to the period between structural breaks that we have identified.
Source: BEA, Haver, Nomura

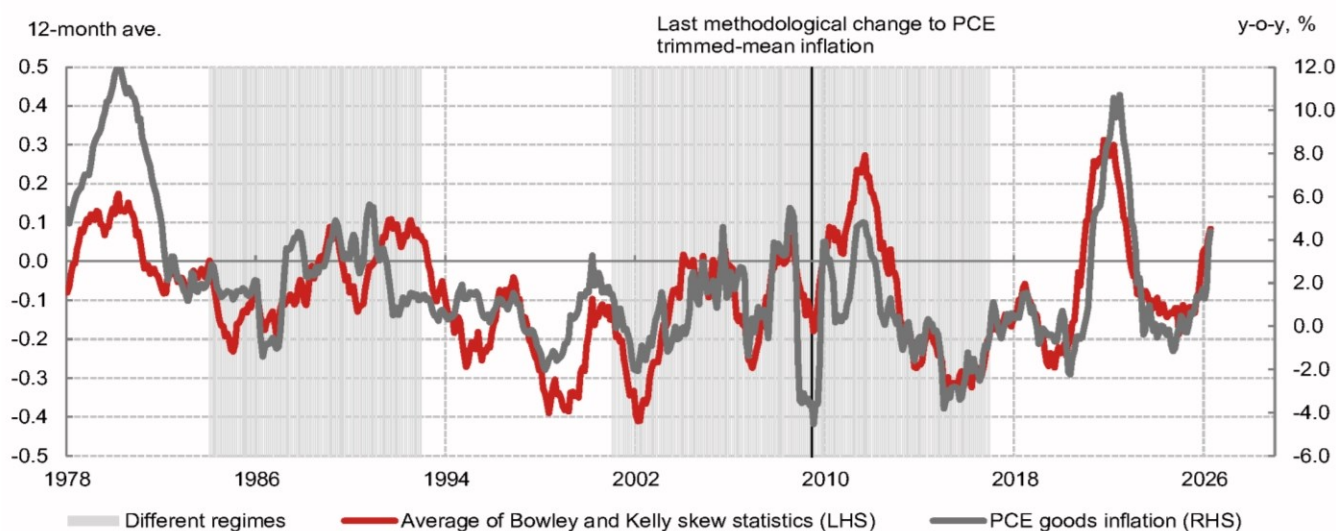
If the distribution became more positively skewed, PCE trimmed-mean inflation would ignore the upper tail of the distribution excessively and understate the underlying inflation. Conversely, in the case of a more negatively skewed distribution, PCE trimmed-mean inflation would overstate the inflation trend.

Since the last methodological update to PCE trimmed-mean inflation in 2009, the price change distribution has become positively skewed more often (*Fig. 7*).

We found four statistically significant structural breaks in distribution skewness (Bowley and Kelly indices) since 1977 (based on the Bai-Perron test) clustering around 1984, 1993, 2001-02, and 2017-19. Notably, these breaks closely coincide with structural breaks in goods inflation, suggesting that shifts in goods price dynamics are likely the primary driver of changes in distribution symmetry (*Fig. 8*).

Fig. 8: We identified four structural breaks in distribution skewness, which closely coincide with structural breaks in goods inflation, suggesting that shifts in goods price dynamics are likely the primary driver of changes in distribution symmetry

Skewness of the cross-sectional price change distribution vs. PCE goods price inflation



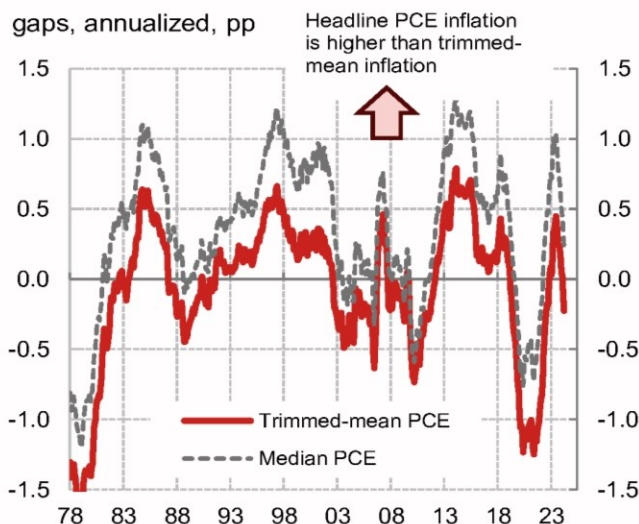
Note: *Regimes refer to the period between structural breaks that we have identified.
Source: BEA, Haver, Nomura

The trimmed-mean measure understates (overstates) the true inflation when the price distribution is positively (negatively) skewed, which means that the negative bias narrowed during the goods deflation period (following China's entry in WTO) and widened when goods inflation is elevated (post-the trade war/pandemic). Overall, the distribution has exhibited positive skewness more frequently since 2010 (especially post-pandemic). This means the upper-tail trim, which was calibrated on 1977-2009 data, is too large, causing PCE trimmed-mean inflation understate the underlying inflation trend.

PCE trimmed-mean inflation appears to understate underlying inflation by 48bp

We attempt to quantify how much PCE trimmed-mean inflation is downwardly biased. We first calculate the monthly gap between annualized PCE trimmed-mean inflation and headline PCE inflation. We then take a 36-month centered moving average of the gap to estimate a "bias" after adjusting for outliers (*Fig. 9*). Following the approach laid out in *Cleveland Fed* note, we estimate how much the bias was caused by the skewness (measured by the average of Bowley and Kelly coefficients of skewness).

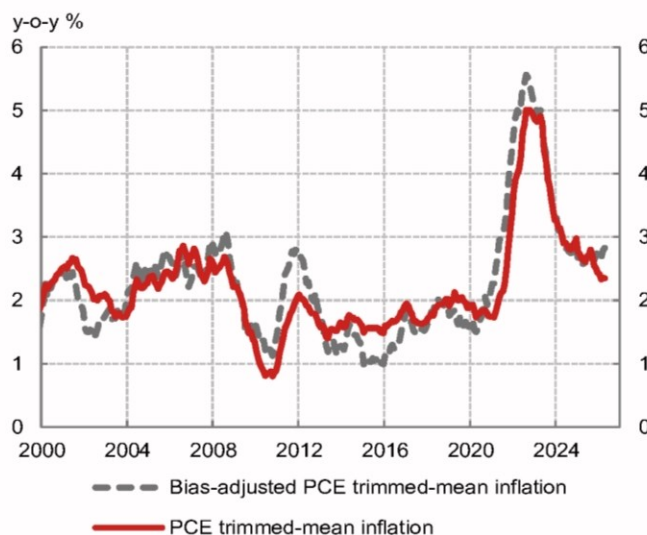
Fig. 9: Gaps between headline PCE inflation and alternative inflation measures



Note: 36-month centered moving averages of the monthly annualized inflation gap from actual headline PCE inflation to estimate an "bias" after adjusting for outliers.
Source: BEA, Haver, Nomura

Fig. 10: PCE trimmed-mean inflation appears to understate the underlying inflation trend

PCE trimmed-mean inflation: official vs. bias-adjusted



Source: BEA, Dallas Fed, Haver, Nomura

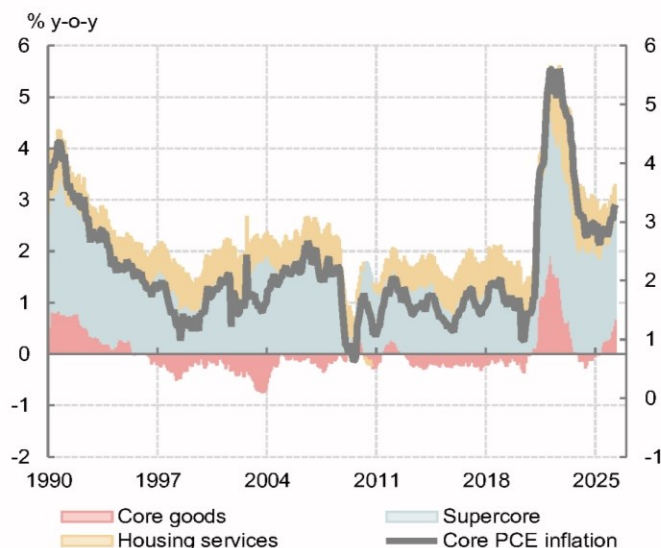
We find that PCE trimmed-mean inflation is currently lower than the bias-adjusted trimmed-mean inflation by about 48bp on a y-o-y basis (*Fig. 10*). This result indicates that the bias-adjusted trimmed-mean rate of inflation stands at 2.8%, narrowing the gap with core PCE inflation of 3.3% as of April 2026. Note that PCE trimmed-mean inflation and its bias-adjusted version moved in the different directions in recent months.

Changing goods price dynamics could complicate the picture

Goods prices are a key driver of the recent shift in the skewness of the inflation distribution. The two measures of skewness are correlated with core goods price inflation, which suggests that more frequent positive goods prices inflation prints appear to have played a key role behind the rightward shift in the cross-sectional distribution of monthly price changes. Core goods price inflation, which used to be a persistent deflationary force, has consistently pushed inflation higher in the post-COVID era (*Fig. 11*).

Fig. 11: Core goods price inflation might no longer be a reliable source of disinflation

Decomposition of core PCE inflation



Source: BEA, Haver, Nomura

AI investment boom likely to keep core goods price inflation positive

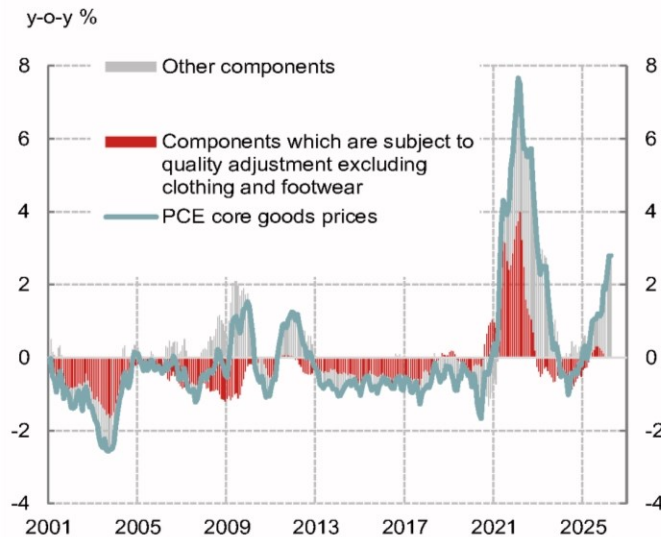
Core goods inflation will probably remain positive in the coming years, which warrants a revamp of PCE trimmed-mean inflation. In particular, the ongoing AI investment boom and resulting upward pressures on key electronic components such as semiconductors could keep boosting core goods prices.

In the 2010s, PCE core goods price inflation was almost persistently negative, led by lower prices of consumer electronics, such as computers, televisions, vehicles, and household appliances. These components are subject to quality adjustments by the BLS/BEA, which means that adding more advanced functions to those products are treated as price declines, all else equal. A large part of core goods deflation in the 2010s came from commodities which are adjusted for quality changes (*Fig. 12*).

Note that goods excluding clothing and footwear which are subject to quality adjustments account for less than 20% of total PCE core goods components. However, many of these tech-heavy products now face upward price pressures due to global shortages of semiconductors, as well as supply-chain disruptions induced by the Iran war (*Fig. 13*). The emerging upward pressures seem to outweigh the persistent negative impact from quality adjustment. If the AI investment boom continues and keeps certain key components for electronics in short supply, core goods price inflation would remain positive for an extended period.

Fig. 12: PCE core goods price inflation was historically weighed down by components subject to quality adjustment

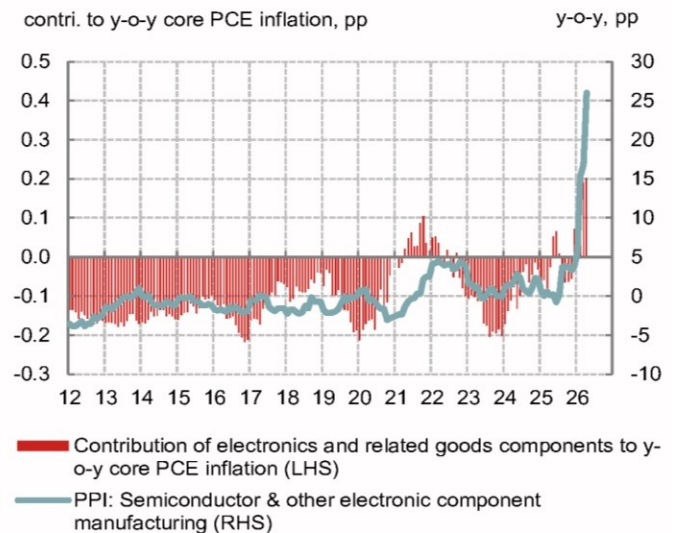
Decomposition of PCE core goods price inflation



Source: BEA, Haver, Nomura

Fig. 13: Higher semiconductor prices pose an upside risk to PCE core goods prices

Contribution of electronics and related goods to core PCE inflation vs. PPI's semiconductor prices



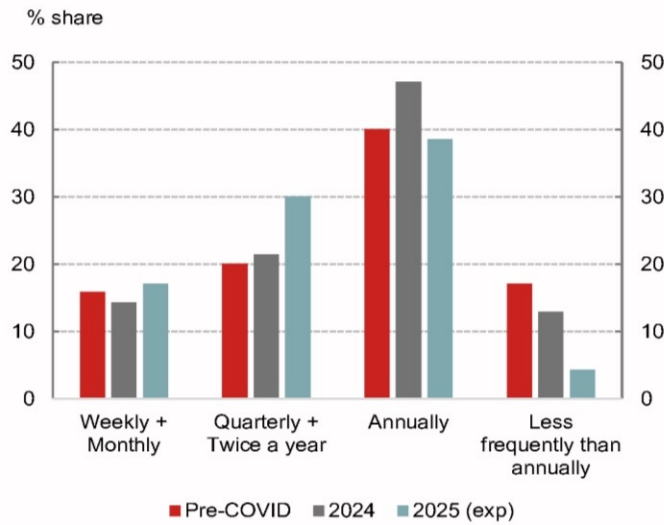
Source: BEA, BLS, Nomura

Changes in businesses' pricing strategy also indicate structural changes in goods price dynamics

Since the pandemic, business practices on pricing appear to have changed. According to a *Richmond Fed survey*, the share of manufacturers changing prices at least once per year is higher than in the pre-pandemic era while the share of firms changing their prices less frequently decreased (*Fig. 14*). In addition, goods prices might have become more susceptible to changes in inventories. Even after the post-pandemic inflation acceleration subsided, the inventory-to-sales ratio and PCE core goods price inflation showed a negative correlation between 2024-2025 (lower inventory-to-sales leads to more frequent price increases) (*Fig. 15*). Conventional wisdom is that core goods prices are determined by global prices and are less responsive to changes in domestic demand-supply balance. However, this might be changing and goods price inflation has become more cyclical.

Fig. 14: Manufacturers change prices more frequently now than in the pre-pandemic era

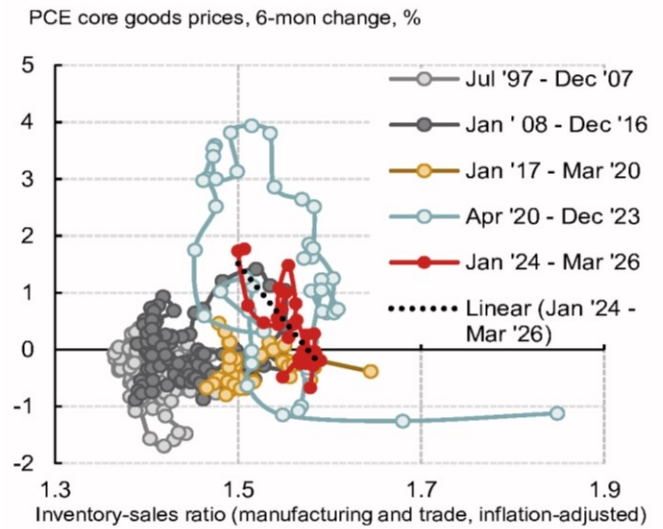
Share of manufacturers by frequency of price changes



Source: Richmond Fed, Nomura

Fig. 15: PCE core goods prices and inventory-sales ratio are more negatively correlated

PCE core goods inflation vs US business inventory-to-sales ratio



Source: BEA, Census Bureau, Haver, Nomura

Policy implications

Dovish policymakers including former Fed Chair Powell tend to see recent inflation acceleration led by higher goods prices as temporary and maintain their sanguine inflation outlook in the medium term. However, persistently positive inflation of core goods prices and a resulting shift in the skewness of cross-sectional inflation distribution undermine the usefulness of certain alternative inflation measures such as PCE trimmed-mean inflation. Monetary policy driven by suboptimal inflation metrics might increase the risk of falling “behind the curve,” leading to more aggressive rate hikes later.

Appendix A-1

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